

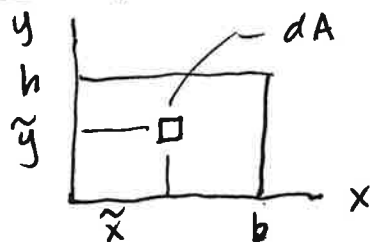
Centroid:

By Integration

$$\bar{x} = \frac{\int \tilde{x} dA}{\int dA}$$

$$\bar{y} = \frac{\int \tilde{y} dA}{\int dA}$$

Rectangle:



Denominator:

$$A = \int_0^h \int_0^b dx dy = \boxed{bh = A}$$

TI-89:

$$A = \int (\int (1, x, 0, b), y, 0, h)$$

$\bar{x}$ Numerator:

$$\int \tilde{x} dA = \int_0^h \int_0^b x dx dy =$$

TI-89

$$= \int (\int (x, x, 0, b), y, 0, h) = \frac{b^2 h}{2}$$

Finally:

$$\bar{x} = \frac{\frac{b^2 h}{2}}{bh} = \boxed{\frac{b}{2} = \bar{x}}$$

$\bar{y}$ Numerator:

$$\int \tilde{y} dA = \int_0^h \int_0^b y dx dy$$

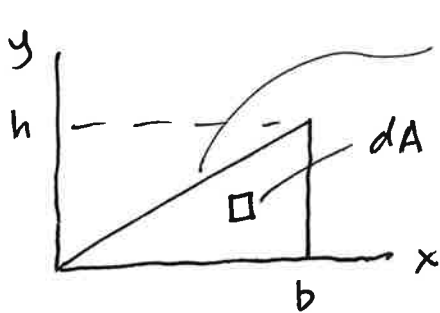
TI-89

$$= \int (\int (y, x, 0, b), y, 0, h) = \frac{bh^2}{2}$$

Finally:

$$\bar{y} = \frac{\frac{bh^2}{2}}{bh} = \boxed{\frac{h}{2} = \bar{y}}$$

Centroid of a Right Triangle (Integration)



Slope: $m = h/b$

$$y = \left(\frac{h}{b}\right)x$$

$$\bar{x} = \frac{\int \tilde{x} dA}{\int dA}$$

$$\bar{y} = \frac{\int \tilde{y} dA}{\int dA}$$

Denominator:

$$A = \int_0^b \int_0^{(h/b)x} dy dx$$

TI-89

$$A = \int \left(\int (1, y, 0, (h/b)*x), x, 0, b \right) = \frac{bh}{2}$$

$\bar{x}$ Numerator:

$$\int \tilde{x} dA = \int_0^b \int_0^{(h/b)x} x dy dx$$

TI-89

$$= \int \left(\int (x, y, 0, (h/b)*x), x, 0, b \right) = \frac{b^2 h}{3}$$

$$\bar{x} = \frac{\frac{b^2 h}{3}}{\frac{bh}{2}} \Rightarrow \boxed{\bar{x} = \frac{2}{3}b}$$

$\bar{y}$ Numerator:

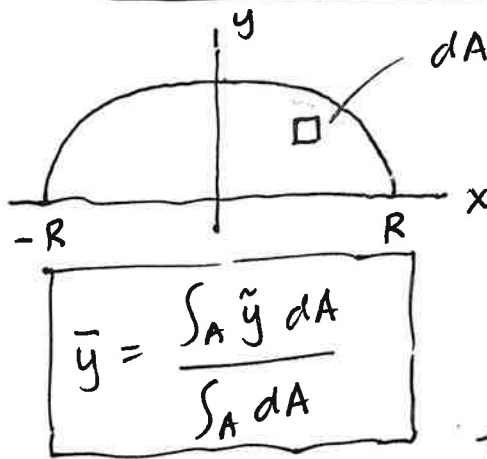
$$\int \tilde{y} dA = \int_0^b \int_0^{(h/b)x} y dy dx$$

TI 89:

$$= \int \left(\int (y, y, 0, (h/b)*x), x, 0, b \right)$$

$$= \frac{bh^2}{6} \quad \bar{y} = \frac{\frac{bh^2}{6}}{\frac{bh}{2}} = \boxed{\frac{1}{3}h = \bar{y}}$$

Centroid of a Semi Circle (by Integration)



$$dA = dx dy$$

$$x^2 + y^2 = R^2$$

$$y = (R^2 - x^2)^{.5}$$

$$\text{Area: } A = \int_{-R}^R \int_0^{(R^2 - x^2)^{.5}} dy dx$$

$$\text{TI-89} = \int (\int (1, y, 0, (r^2 - x^2)^{.5}), x, -r, r)$$

$$A = \frac{\pi r^2}{2}$$

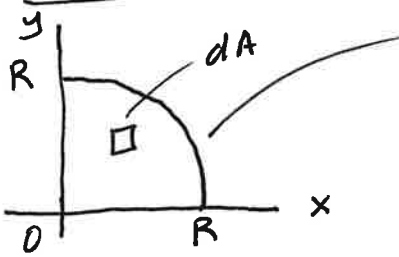
$$\text{Numerator: } \int_{-R}^R \int_0^{(R^2 - x^2)^{.5}} y dy dx$$

$$= \int (\int (y, y, 0, (r^2 - x^2)^{.5}), x, -r, r)$$

$$= \frac{2}{3} r^3$$

$$\text{Finally: } \bar{y} = \frac{\frac{2}{3} r^3}{\frac{\pi r^2}{2}} = \boxed{\frac{4R}{3\pi} = \bar{y}}$$

Centroid of a Quarter Circle (by Integration)



$$x^2 + y^2 = R^2$$

$$y = (R^2 - x^2)^{.5}$$

$$dA = dx dy$$

$$\bar{x} = \frac{\int_A \tilde{x} dA}{\int_A dA}$$

$$\bar{y} = \frac{\int_A \tilde{y} dA}{\int_A dA}$$

Area:

$$A = \int_0^R \int_0^{(R^2 - x^2)^{.5}} dy dx$$

TI-89

$$= \int (\int (1, y, 0, (r^2 - x^2)^{.5}), x, 0, r)$$

$$A = \frac{\pi r^2}{4}$$

$\bar{x}$ Numerator

$$= \int_0^r \int_0^{(r^2 - x^2)^{.5}} x dy dx$$

TI-89

$$= \int (\int (x, y, 0, (r^2 - x^2)^{.5}), x, 0, r)$$

$$= \frac{r^3}{3}$$

$$\bar{x} = \frac{\frac{1}{3} r^3}{\frac{\pi r^2}{4}} = \boxed{\frac{4R}{3\pi} = \bar{x}}$$

$\bar{y}$ Numerator

$$= \int_0^r \int_0^{(r^2 - x^2)^{.5}} y dy dx$$

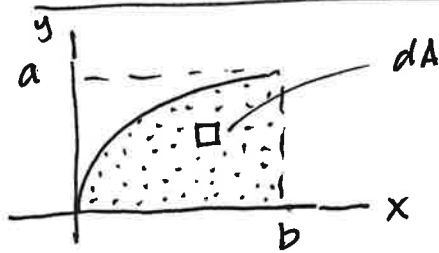
TI-89

$$= \int (\int (y, y, 0, (r^2 - x^2)^{.5}), x, 0, r)$$

$$= \frac{r^3}{3}$$

$$\bar{y} = \frac{\frac{1}{3} r^3}{\frac{\pi r^2}{4}} = \boxed{\frac{4R}{3\pi} = \bar{y}}$$

Centroid of a Parabola (by Integration)



a parabola on its side has the functional form:

$$y^2 = cx \quad \text{or} \quad y = \sqrt{cx}$$

To find c : when $x=b$, $y=a$

$$a^2 = c \cdot b \Rightarrow \boxed{c = a^2/b}$$

Finally, $\boxed{y^2 = (a^2/b)x}$

or

$$\boxed{y = \sqrt{(a^2/b)x}}$$

Area: $A = \int_A dA = \int_0^b \int_0^{\sqrt{(a^2/b)x}} dy dx$

TI-89 = $\int(\int(1, y, 0, ((a^2/b)*x)^{.5}), x, 0, b)$

$$A = \frac{2a(\frac{1}{b})^{3/2} \cdot b^{5/2}}{3} = \boxed{\frac{2}{3}ab = A}$$

$\bar{x}$ Numerator: $= \int_0^b \int_0^{\sqrt{(a^2/b)x}} x dy dx$

TI-89 = $\int(\int(x, y, 0, ((a^2/b)*x)^{.5}), x, 0, b)$

$$= \frac{2ab^2}{5}$$

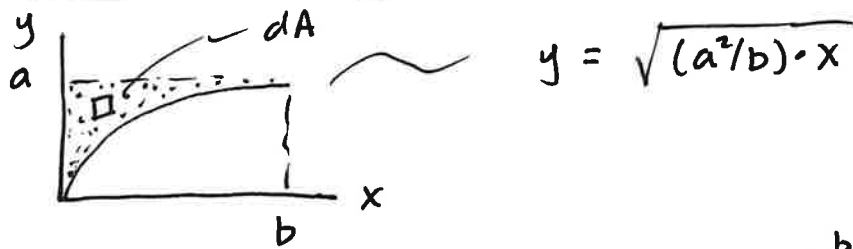
$$\bar{x} = \frac{\frac{2ab^2}{5}}{\frac{2}{3}ab} \Rightarrow \boxed{\bar{x} = \frac{3}{5}b}$$

$\bar{y}$ Numerator: $= \int_0^b \int_0^{\sqrt{(a^2/b)x}} y dy dx$

TI-89 = $\int(\int(y, y, 0, ((a^2/b)*x)^{.5}), x, 0, b)$

$$= \frac{a^2 \cdot b}{4} \quad \bar{y} = \frac{\frac{a^2 \cdot b}{4}}{\frac{2}{3}ab} = \boxed{\frac{3}{8}a = \bar{y}}$$

Centroid of a Spandrel (by Integration)



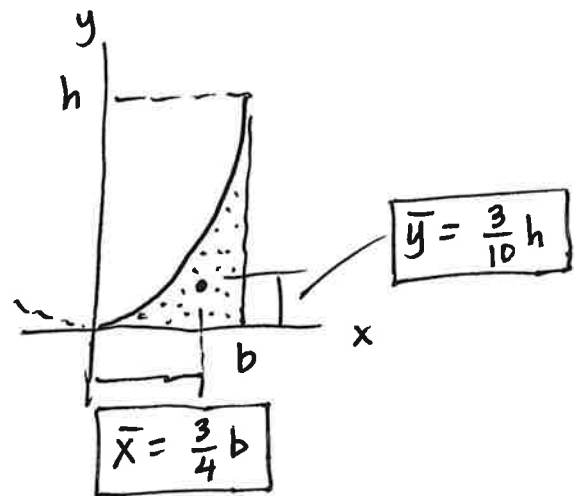
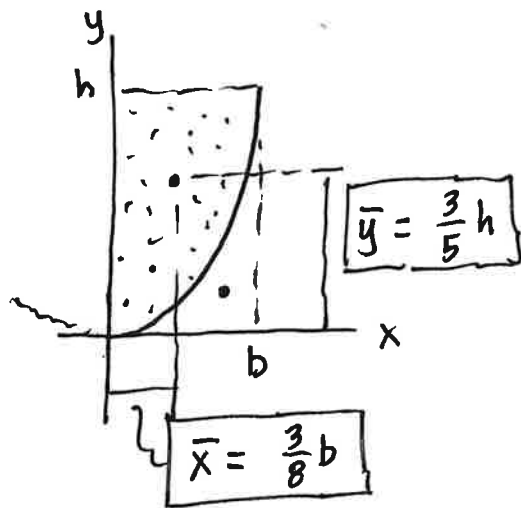
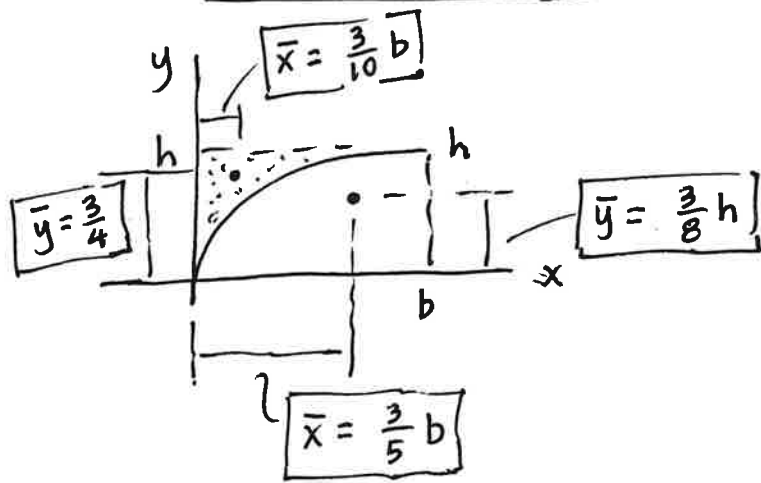
Area: $A = \int_A dA = \int_0^b \int_{((a^2/b) \cdot x)^{.5}}^a dy dx$

TI-89 $= \int(\int(1, y, ((a^2/b) * x)^{.5}, a), x, 0, b)$
 $= ab - \frac{2a(\frac{1}{b})^{3/2} \cdot b^{5/2}}{3}$
 $= ab - \frac{2ab}{3} = ab(1 - \frac{2}{3}) = \boxed{\frac{1}{3}ab = A}$

$\bar{x}$ Numerator: $= \int_0^b \int_{((a^2/b) \cdot x)^{.5}}^a x dy dx$
 $= \int(\int(x, y, ((a^2/b) * x)^{.5}, a), x, 0, b)$
 $= \frac{ab^2}{10}$
Finally, $\bar{x} = \frac{\frac{ab^2}{10}}{\frac{1}{3}ab} \Rightarrow \boxed{\bar{x} = \frac{3}{10}b}$

$\bar{y}$ Numerator: $= \int(\int(y, y, ((a^2/b) * x)^{.5}, a), x, 0, b)$
 $= \frac{a^2b}{4}$
 $\bar{y} = \frac{\frac{a^2b}{4}}{\frac{1}{3}ab} \Rightarrow \boxed{\bar{y} = \frac{3}{4}a}$

Centroids of a Parabola



$$\bar{X} = \frac{\sum \tilde{y} A}{\sum A}$$

$$= \frac{(\frac{b}{2})(bh) - \frac{3}{8}(b)(\frac{2}{3}bh)}{bh - \frac{2}{3}bh}$$

$$\bar{X} = \frac{\frac{b}{2} - \frac{1}{4}b}{1/3} = \frac{\frac{1}{4}b}{1/3} = \boxed{\frac{3}{4}b = \bar{X}}$$

$$\bar{y} = \frac{\sum \tilde{x} A}{\sum A}$$

$$= \frac{(\frac{h}{2})(bh) - (\frac{3}{5}h)(bh)(\frac{2}{3})}{\frac{1}{3}bh}$$

$$= \frac{h(\frac{1}{2} - \frac{2}{5})}{1/3} = \frac{\frac{5}{10} - \frac{4}{10}}{1/3} = \boxed{\frac{3}{10}h = \bar{y}}$$